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ENGS 93 Homework - Part 2

Confidence Intervals and Hypothesis Testing Dataset: Engineering Salaries 2021

This analysis examines the distribution of engineering base salaries and applies statistical inference to estimate and test the population mean. The analysis uses a sample of 10,000 observations.

Question 1 evaluates Normality and the adequacy of the sample size. Question 2 constructs confidence intervals for the population mean. Question 3 performs a two-sided hypothesis test.

```
clear;
clc;
close all;
```

Data Preparation

The Engineering Salaries 2021 dataset contains salary information, professional experience, and employment characteristics for 10,000 observations.

We use annual base salary as our continuous variable of interest. This maintains consistency with Part 1 and allows us to examine statistical inference using a large sample.

```
% the repository root is specified explicitly because matlab's
% publishing process may execute scripts from a temporary directory.
project_root = '/Users/ritisingh/Desktop/engs93_homework';

% the dataset is stored in the repository's shared data directory.
data_path = fullfile(project_root, 'data', ...
    'Engineering_Salaries_2021_10000.csv');

% the dataset must be present before the statistical analysis can run.
assert(isfile(data_path), ...
    'The salary dataset was not found in the data directory.');

% the csv file is imported as a matlab table with its original headers.
data = readtable(data_path);
```

```
% annual base salary is the continuous variable used in this analysis.
salary = data.base_salary;

% non-finite observations are excluded to prevent invalid calculations.
salary = salary(isfinite(salary));
salary = salary(:);

% the sample size, mean, and standard deviation describe the data.
n = length(salary);
salary_mean = mean(salary);
salary_std = std(salary);

% the estimated standard error describes the sampling variability
% of the mean under appropriate independence assumptions.
standard_error = salary_std / sqrt(n);

% the relevant sample statistics are shown before the analysis.
fprintf('\nPART 2: SAMPLE STATISTICS\n');
fprintf('Observations: %d\n', n);
fprintf('Mean base salary: $%.2f\n', salary_mean);
fprintf('Sample standard deviation: $%.2f\n', salary_std);
fprintf('Estimated standard error: $%.2f\n', standard_error);
```

```
PART 2: SAMPLE STATISTICS
Observations: 10000
Mean base salary: $135574.20
Sample standard deviation: $60185.29
Estimated standard error: $601.85
```

Question 1: Normality and Sample Size

We assess whether annual base salary is approximately Normally distributed using a normalized histogram and a Normal Q-Q plot.

The histogram shows the shape of the observed distribution relative to a Normal density curve fitted using the sample mean and standard deviation. A Normal distribution is symmetric and bell-shaped.

The Q-Q plot compares observed salary quantiles with theoretical quantiles from the fitted Normal distribution. If the distributions were similar, the plotted points would follow the diagonal reference line with relatively small deviations.

Question 1: Histogram and Fitted Normal Density

```
% the salary values are distributed across a smooth vector so that
% the theoretical normal density can be evaluated continuously.
x = linspace(min(salary), max(salary), 1000);

% the normal density uses the observed sample mean and standard
% deviation rather than a predetermined population distribution.
```

```
normal_pdf = (1 / (salary_std * sqrt(2*pi))) .* ...
    exp(-0.5 * ((x - salary_mean) ./ salary_std).^2);

% a white figure and axes background improve readability when the
% graph is included in a published pdf or printed report.
fig_hist = figure('Color', 'w', ...
    'Position', [100, 100, 850, 500]);

ax_hist = axes('Parent', fig_hist);
set(ax_hist, 'Color', 'w', ...
    'XColor', 'k', 'YColor', 'k', ...
    'FontSize', 11);

% probability-density normalization makes the histogram comparable
% to the fitted normal probability density function.
histogram(ax_hist, salary, 40, ...
    'Normalization', 'pdf', ...
    'FaceColor', [0.25, 0.45, 0.70], ...
    'FaceAlpha', 0.75, ...
    'EdgeColor', 'w');

hold(ax_hist, 'on');

% the fitted normal curve illustrates how closely a symmetric
% theoretical distribution represents the observed salary values.
plot(ax_hist, x, normal_pdf, ...
    'Color', [0.80, 0.20, 0.20], ...
    'LineWidth', 2.2);

xlabel(ax_hist, 'Annual base salary ($)');
ylabel(ax_hist, 'Probability density');
title(ax_hist, ...
    'Figure 1. Distribution of Engineering Base Salaries');

legend(ax_hist, ...
    'Observed salary distribution', ...
    'Fitted Normal distribution', ...
    'Location', 'northeast');

% the displayed axis focuses on the central portion of the salary
% distribution. all observations remain in the statistical analysis.
xlim(ax_hist, [0, 400000]);

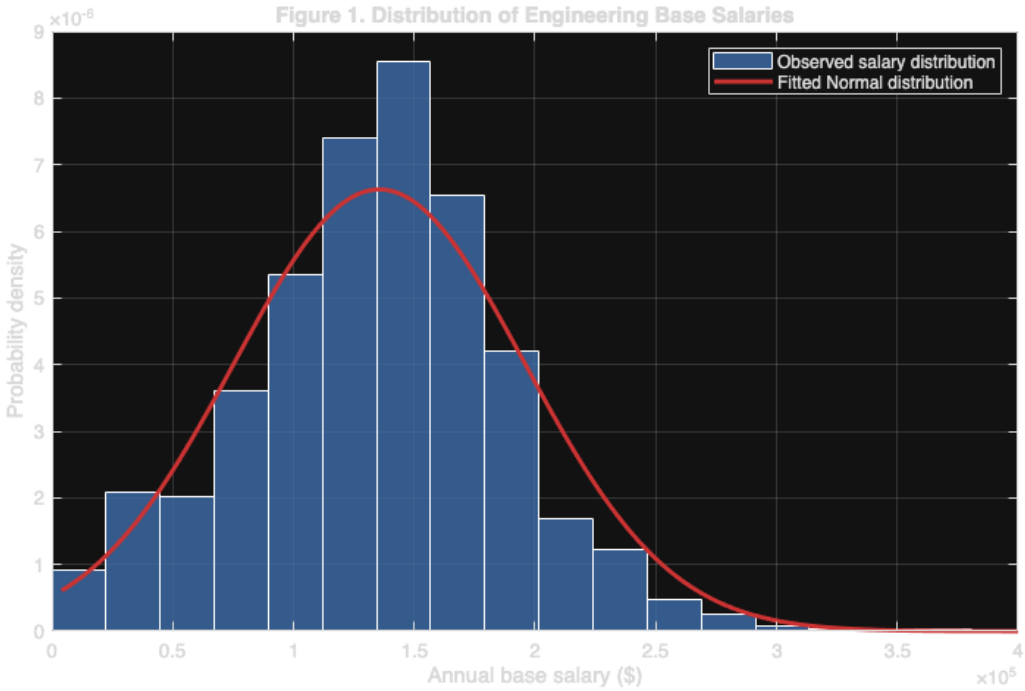
grid(ax_hist, 'on');
ax_hist.GridAlpha = 0.15;

hold(ax_hist, 'off');

% the image is saved directly without reopening the figure later.
results_dir = fullfile(project_root, 'results', 'figures');

if ~isfolder(results_dir)
    mkdir(results_dir);
end
```

```
exportgraphics(fig_hist, ...
    fullfile(results_dir, 'part2_salary_histogram.png'), ...
    'Resolution', 300);
```



Question 1: Normal Q-Q Plot

```
% the empirical salary quantiles are obtained by ordering the
% observations from smallest to largest.
sorted_salary = sort(salary);

% plotting positions associate each ordered observation with
% a cumulative probability between zero and one.
probabilities = ((1:n)' - 0.5) / n;

% the inverse normal distribution is calculated using erfinv,
% avoiding a dependency on the statistics toolbox.
theoretical_quantiles = salary_mean + salary_std * ...
    sqrt(2) .* erfinv(2 * probabilities - 1);

% the q-q plot uses the same visual styling as the histogram.
fig_qq = figure('Color', 'w', ...
    'Position', [100, 100, 850, 500]);

ax_qq = axes('Parent', fig_qq);
set(ax_qq, 'Color', 'w', ...
    'XColor', 'k', 'YColor', 'k', ...
    'FontSize', 11);
```

```
% every point compares an observed salary quantile against the
% corresponding theoretical quantile under the fitted normal model.
plot(ax_qq, theoretical_quantiles, sorted_salary, '.', ...
    'Color', [0.25, 0.45, 0.70], ...
    'MarkerSize', 7);

hold(ax_qq, 'on');

% the diagonal reference line represents equality between observed
% and theoretical quantiles on the same salary scale.
reference_min = min(theoretical_quantiles);
reference_max = max(theoretical_quantiles);

plot(ax_qq, ...
    [reference_min reference_max], ...
    [reference_min reference_max], ...
    'Color', [0.80, 0.20, 0.20], ...
    'LineWidth', 2);

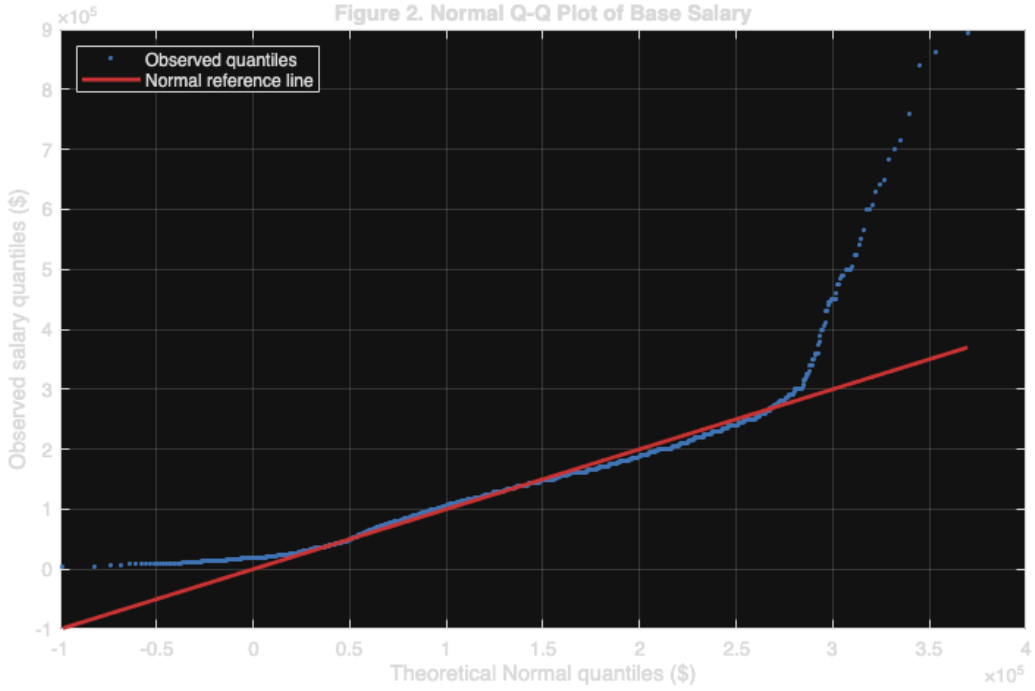
xlabel(ax_qq, 'Theoretical Normal quantiles ($)');
ylabel(ax_qq, 'Observed salary quantiles ($)');
title(ax_qq, 'Figure 2. Normal Q-Q Plot of Base Salary');

legend(ax_qq, ...
    'Observed quantiles', ...
    'Normal reference line', ...
    'Location', 'northwest');

grid(ax_qq, 'on');
ax_qq.GridAlpha = 0.15;

hold(ax_qq, 'off');

% the second figure is also saved at publication quality.
exportgraphics(fig_qq, ...
    fullfile(results_dir, 'part2_salary_qqplot.png'), ...
    'Resolution', 300);
```



Question 1: Interpretation

The histogram indicates that engineering base salaries are positively skewed rather than symmetric and bell-shaped. Although the majority of observations are concentrated near the middle of the distribution, some substantially higher salaries produce an extended right tail.

The Normal Q-Q plot supports this conclusion. The observed quantiles deviate systematically from the theoretical reference line, most noticeably in the upper tail. These deviations indicate that the individual salary observations are not approximately Normally distributed across the full range of values.

However, the sample contains 10,000 observations. Under suitable independence and finite-variance assumptions, the Central Limit Theorem implies that the sampling distribution of the sample mean approaches a Normal distribution as the sample size increases.

Therefore, despite the non-Normal distribution of individual salaries, this sample size supports using a large-sample z approximation for inference about the population mean.

This conclusion assumes that observations are sufficiently independent and that the dataset reasonably represents the population of interest. A large sample does not eliminate sampling bias or dependence between observations.

Note: Figure 1 restricts the horizontal display to $0-$400,000 for readability. All salary observations, including higher values, are included in the statistical calculations and fitted distribution.

Question 2: Confidence Intervals for the Population Mean

We construct 95% and 75% confidence intervals for the population mean engineering base salary.

Each interval is centered on the observed sample mean, with a margin of error obtained by multiplying the estimated standard error by the appropriate critical value from the standard Normal distribution.

The population standard deviation is unknown, so the sample standard deviation is used to estimate the standard error. Given the large sample size, this provides a reasonable Normal approximation.

A higher confidence level corresponds to a larger critical value, resulting in a wider confidence interval.

```
% the two-sided critical values correspond to the central
% 95% and 75% areas of the standard normal distribution.
z_95 = sqrt(2) * erfinv(0.95);
z_75 = sqrt(2) * erfinv(0.75);

% the confidence interval margins account for the estimated
% sampling variability of the sample mean.
margin_95 = z_95 * standard_error;
margin_75 = z_75 * standard_error;

% both intervals are centered on the same observed sample mean.
ci_95 = [salary_mean - margin_95, ...
         salary_mean + margin_95];

ci_75 = [salary_mean - margin_75, ...
         salary_mean + margin_75];

fprintf('\nQUESTION 2: CONFIDENCE INTERVALS\n');

fprintf('95%% confidence interval: $%.2f to $%.2f\n', ...
    ci_95(1), ci_95(2));

fprintf('95%% margin of error: $%.2f\n', margin_95);

fprintf('\n75%% confidence interval: $%.2f to $%.2f\n', ...
    ci_75(1), ci_75(2));

fprintf('75%% margin of error: $%.2f\n', margin_75);
```

```
QUESTION 2: CONFIDENCE INTERVALS
95% confidence interval: $134394.59 to $136753.81
95% margin of error: $1179.61

75% confidence interval: $134881.86 to $136266.54
75% margin of error: $692.34
```

Question 2: Interpretation

The 95% confidence interval for the population mean base salary is approximately $134,394.59 to $136,753.81, while the 75% confidence interval is approximately $134,881.86 to $136,266.54.

Both intervals are centered on the sample mean of $135,574.20. The 95% interval is wider because its higher confidence level requires a larger margin of error.

Under repeated sampling, the procedure used to construct 95% confidence intervals would capture the true population mean in approximately 95% of samples, assuming the statistical conditions are satisfied. The corresponding coverage rate for the 75% confidence interval procedure would be approximately 75%.

These intervals estimate the population mean, not the range of individual engineering salaries. Their relatively narrow widths reflect the precision associated with the large sample size.

However, precision does not guarantee representativeness. If the dataset is systematically biased or contains dependent observations, the intervals may not achieve their nominal coverage for the broader population of engineering professionals.

Question 3: Hypothesis Test for Mean Base Salary

We test whether the population mean engineering base salary differs from an illustrative reference value of $130,000.

The null hypothesis states that the population mean is $130,000:

H0: mu = 130000

The alternative hypothesis states that the mean is not $130,000:

HA: mu ~= 130000

Because deviations in either direction are relevant, a two-sided test is appropriate. We use a significance level of alpha = 0.05.

The test statistic compares the sample mean against the null value in units of the estimated standard error.

```
% the selected benchmark is illustrative rather than an externally
% verified average salary for the engineering population.
mu_0 = 130000;

% the significance level determines the rejection threshold
% for the statistical decision.
alpha = 0.05;

% the observed z-statistic measures the standardized difference
% between the sample mean and the hypothesized population mean.
z_stat = (salary_mean - mu_0) / standard_error;

% the complementary error function evaluates the two-sided
% standard normal tail probability without a statistics toolbox.
p_value = erfc(abs(z_stat) / sqrt(2));

% the null hypothesis is rejected when the p-value is below alpha.
reject_null = p_value < alpha;

fprintf('\nQUESTION 3: HYPOTHESIS TEST\n');

fprintf('Null hypothesis: mean salary = $%.2f\n', mu_0);
fprintf('Alternative: mean salary differs from $%.2f\n', mu_0);
fprintf('Observed sample mean: $%.2f\n', salary_mean);
```

```
fprintf('Z-statistic: %.4f\n', z_stat);
fprintf('Two-sided p-value: %.6g\n', p_value);
fprintf('Significance level: %.2f\n', alpha);

if reject_null
    fprintf('Decision: Reject the null hypothesis.\n');
else
    fprintf('Decision: Fail to reject the null hypothesis.\n');
end
```

```
QUESTION 3: HYPOTHESIS TEST
Null hypothesis: mean salary = $130000.00
Alternative: mean salary differs from $130000.00
Observed sample mean: $135574.20
Z-statistic: 9.2617
Two-sided p-value: 2.01143e-20
Significance level: 0.05
Decision: Reject the null hypothesis.
```

Question 3: Interpretation

The observed sample mean is $135,574.20, which exceeds the hypothesized population mean of $130,000 by $5,574.20.

The resulting z-statistic is approximately 9.2617, with a two-sided p-value of approximately 2.01 x 10^(-20).

Since the p-value is substantially smaller than the significance level of 0.05, we reject the null hypothesis. Under the assumptions of the test, the sample provides strong statistical evidence that the population mean base salary differs from $130,000.

Although the sample mean is higher than the hypothesized value, the test is two-sided and formally evaluates whether the mean differs in either direction.

Statistical significance does not necessarily imply practical significance. With 10,000 observations, even a difference that is relatively modest compared with overall salary variability can produce a very small p-value.

Additionally, the population standard deviation is not known. This analysis therefore uses a large-sample z approximation with the sample standard deviation in the estimated standard error, rather than an exact z-test with a known population standard deviation.

The interpretation also depends on the independence and representativeness of the salary observations.

Part 2: Conclusion

Part 2 examined the distribution of engineering base salaries and applied statistical inference to the population mean.

The histogram and Q-Q plot indicate that individual salaries are positively skewed and do not closely follow a Normal distribution. Nevertheless, the large sample supports approximate Normal inference about the sample mean under suitable assumptions.

The 95% and 75% confidence intervals provide different levels of confidence in the population mean estimate, demonstrating the relationship between confidence level and interval width.

The two-sided hypothesis test provides strong evidence that the population mean base salary differs from the illustrative benchmark of $130,000.

These results demonstrate how graphical analysis, confidence intervals, and hypothesis testing can be combined to evaluate statistical properties of an engineering dataset.

```
fprintf('\nPart 2 analysis complete.\n');
```

```
Part 2 analysis complete.
```

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